PRASHANTH KUMAR SIRUSALA

Data Engineer
786-561-6223 · prashanthsirusala2209@gmail.com · Miami, FL · linkedin.com/in/psirusala · Portfolio

SUMMARY

Data Engineer with 5 years owning AWS and Snowflake data pipelines end to end, from ingestion and data modeling to production on-call, on security-event, healthcare and advertising data. Handled a 17B-row Snowflake workload, Redshift reporting tables of up to 450M rows and nightly SSIS loads from 30+ healthcare source systems. Built pipelines in SQL, Python, PySpark and dbt on Airflow and Step Functions, and led a team of two data engineers working directly with analysts and business stakeholders.

TECHNICAL SKILLS

Languages & Processing: SQL, Python, PySpark (Apache Spark), Snowpark, dbt, Jinja, Airflow/MWAA, SSIS, Informatica, Fivetran, Apache Kafka, REST and GraphQL APIs, SFTP, ETL/ELT, CDC, window functions

Warehousing & Modeling: Snowflake, Amazon Redshift, PostgreSQL, SQL Server, Oracle, DB2, Apache Iceberg, data warehousing, dimensional data modeling, star schemas, materialized views, data quality and reconciliation checks

AWS: Glue, Glue Data Catalog, Lambda, Step Functions, S3, Athena, Kinesis, EventBridge, SQS, SNS, DMS, AppFlow, Aurora, DynamoDB, Redshift Data API, IAM, KMS, Secrets Manager, CloudWatch

DevOps & Monitoring: Terraform, AWS CDK, Git, GitHub Actions, Bitbucket, CI/CD, Docker, Kubernetes, Snyk, Datadog

Security & Governance: HIPAA, GDPR, CCPA, PHI/PII handling, Snowflake RBAC, data masking, data retention, lineage

GenAI & BI: Claude and GPT APIs, Snowflake Cortex (LLM functions), prompt-driven SQL generation, Snowflake MCP servers, Streamlit, Tableau, Sigma, Power BI, Power Query, Salesforce Marketing Cloud

Delivery & Operations: production support, on-call, troubleshooting, root-cause analysis, performance tuning, stakeholder communication, requirements gathering, cross-functional collaboration, team leadership, technical documentation

EXPERIENCE

Data Engineer, Tech Detectives Inc., Miami, FL February 2025 – Present

- Delivered AWS and Snowflake data engineering for consulting clients, shipping 2–3 new production pipelines every Agile sprint across batch, streaming (Amazon Kinesis, Apache Kafka), event-driven, API and CDC workloads.
- Cut average query time from 8 minutes to 1–2 minutes on a 17B-row Snowflake security-event workload: traced it to repeated raw-table scans and semi-structured parsing, then moved high-use attributes into incrementally refreshed relational tables.
- Modernized legacy SQL Server and SSIS reporting into an S3, Apache Iceberg and Athena lakehouse fed by CDC and orchestrated by Step Functions, cutting the refresh window from 18 hours to 2 hours.
- Built two dbt projects of about 150 models each on a medallion (bronze, silver, gold) architecture, one on Snowflake and one on Redshift orchestrated by Airflow/MWAA, with incremental models, dimensional data modeling and data quality checks.
- Designed reusable PySpark and SQL ingestion patterns for SaaS and API sources with watermarked incremental landing.
- Built an event-driven file-processing architecture on AWS: SQS queues buffered incoming source files, Step Functions routed each file to a source-specific Glue job, and a completion gate held fact loads until every required feed landed.
- Handled 1–2 on-call requests a day for client production pipelines: triaged CloudWatch and Datadog monitoring alerts, traced failed Glue jobs, Step Functions runs and Snowflake loads to root cause and fixed most within 2–3 hours.
- Added an AI-assisted ETL coverage gate to CI/CD: on each pull request to stg, GitHub Actions, Lambda and the Claude API traced raw-to-curated coverage in dev, posted a report on the PR and flagged anything under 90%.
- Set up Snowflake MCP servers and a chatbot over governed data so business users could query it in plain English, cutting ad hoc data questions to the engineering team from about 10 a week to 2.
- Led a team of two data engineers while staying hands-on, owning delivery of our AWS and Snowflake client work.
- Partnered directly with cross-functional analysts and business stakeholders to gather reporting requirements, set priorities, and communicate delivery status, incidents and trade-offs throughout each engagement.
- Managed client infrastructure as code in Terraform and AWS CDK and enforced data governance: IAM/KMS controls, Snowflake RBAC, lineage and freshness checks, and least-privilege access to PII.
- Assessed Redshift Serverless (via Zero-ETL or DMS CDC) against Snowflake for a 67-table Aurora PostgreSQL source with a 1 TB footprint; recommended Snowflake to fit the client's Sigma BI, then modeled the data and built its dbt models.
- Documented the AWS and Snowflake architecture, pipeline data flows, data models and production support procedures.

Data Engineer, Triangulum Insights, Remote January 2024 – February 2025

- Developed and maintained the AWS data platform for an advertising and consumer panel-data business on S3, Glue, Step Functions, Lambda and Redshift, carrying source data from landing to reporting-ready data warehouse tables.
- Managed the 15-table Redshift gold layer behind Tableau reporting, 100–450M rows per table, with incremental loads and business-key deduplication that kept only the latest valid record per key.
- Performance-tuned long-running Redshift procedures: diagnosed workload-management pressure with system tables, then re-sequenced procedure runs and materialized-view refreshes around observed usage patterns.
- Automated API-based intake of external reporting and CRM data into S3 and Redshift with Python.

- Turned reporting-team grain, source and refresh requirements into Redshift table designs and the Glue and Step Functions pipelines behind them, and kept stakeholders informed during data issues and reprocessing.
- Delivered curated Redshift datasets to Tableau reporting users and supported more than 50 Salesforce Marketing Cloud automations covering scheduled refreshes, query activities and campaign sends.
- Troubleshot production pipeline failures to root cause: isolated malformed files, fixed schema drift and metadata issues.
- Documented the Redshift table designs, pipeline architecture and load schedules for the reporting team.

Data Engineer, Mindtree, India January 2020 – May 2022

- Developed and maintained nightly SSIS batch integrations that moved 1–2M rows of healthcare data from 30+ source systems through SQL Server raw and staging layers into standardized reporting tables and views.
- Built a HIPAA- and GDPR-aligned retention pipeline that moved records past the 10-year policy out of healthcare source databases into an AWS archive, keeping 100M+ in-policy rows in each database.
- Gathered the retention rules from business stakeholders, agreeing which healthcare records had to be kept and which could be purged, then implemented them as the pipeline's purge and archive logic.
- Supported 15–20 nightly SSIS packages in production: reconciled source-to-target row counts before publication, traced failed jobs to root cause and escalated source-system delays to upstream teams.
- Applied PHI/PII masking, exclusion and access controls across the nightly healthcare pipelines.
- Documented SSIS package designs, source-to-target mappings and nightly support procedures for the healthcare workflows.

Software Engineer Intern, Mindtree, India June 2019 – December 2019

- Trained in data engineering and SSIS on SQL Server, building ETL packages with control flows, data flow transformations, lookups and error handling to ingest, cleanse and load source records.
- Monitored scheduled ETL batch jobs on call through SQL Server Agent, catching failed or delayed loads early and escalating them to the data engineering team before downstream reporting users were affected.
- Provided production support alongside senior engineers: investigated ETL job failures and data issues through job history and error logs, then reran the affected jobs once the cause was fixed.
- Resolved change requests (CRs) on SSIS packages, SQL queries and source-to-target mappings, testing each change in development and documenting it before its release to production.

CERTIFICATIONS

- AWS Certified Solutions Architect – Associate · CompTIA Security+
- In progress: SnowPro Core, AWS Certified Data Engineer – Associate

EDUCATION

Master of Science (M.S.), Cybersecurity, Florida International University, Miami, FL

Bachelor of Technology (B.Tech), Computer Science & Engineering, Rajiv Gandhi University of Knowledge Technologies India